Student Answer Script Transcription (High-Resolution Stitched Document)

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Solution

Question 1

Given, ___MATH_BLOCK_0___ ___MATH_BLOCK_1___ Since ___$MATH_BLOCK_2$___ is not in S, the zero matrix is not in S.

2. **If $x \in S$ and $y \in S$, then $x + y \in S$:** Let $x =$ ___$MATH_BLOCK_3$___ and $y =$ ___$MATH_BLOCK_4$___ ___MATH_BLOCK_5___ Since -0.5 is not in S, $x + y \neq S$.

3. **If $x \in S$ and $y \in S$, then $xy \in S$:** Let $x =$ ___$MATH_BLOCK_6$___ and $y =$ ___$MATH_BLOCK_7$___ ___MATH_BLOCK_8___ Since $xy \notin S$, condition 3 is not satisfied.

Therefore, S is not a subspace of $M_n(\mathbb{R})$.

Solution

Question 1

The given matrix is:

___MATH_BLOCK_9___

To find the basis for the null space of A, we need to perform row reduction on the matrix.

Step-by-step explanation:

1. **Row Reduction:** - Subtract $\frac{3}{7}$ times row 1 from row 2. - Subtract $\frac{9}{7}$ times row 1 from row 3. - Add $\frac{6}{7}$ times row 1 to row 4.

This results in:

___MATH_BLOCK_10___

2. **Identify the Basis:** - The resulting matrix has only one non-zero row, which means that the null space of A is spanned by the first column vector.

Therefore, the basis for the null space of A is:

___MATH_BLOCK_11___

So, the answer is:

___MATH_BLOCK_12___